

Future Production-Scale Architecture

While the current implementation focuses on a pixel-perfect, frontend-only strategy, this document outlines the future scaling strategy required to transition this clone into a full production-scale vacation-rental marketplace.

Full-Scale System Architecture Diagram

Component Breakdown & Scaling Strategy

1. Frontend & Deployment

Tech Stack Expansion: Transition to Next.js for Server-Side Rendering (SSR).

Scaling: Utilize AWS CloudFront to serve static assets globally.

2. Backend Microservices

Property & Booking Services: Dedicated Go microservices to independently scale high-read vs high-write operations.

Search & Discovery: A dedicated service querying Elasticsearch.

3. Data & Storage Layer

PostgreSQL: Primary relational database ensuring ACID compliance.

Redis: High-speed caching for frequently accessed listings.

Elasticsearch: Dedicated search index optimized for geospatial queries.

AWS S3: Scalable object storage for hosting property images.